

Measurement of Capacitance by Schering Bridge

AIM:

To Determine the Capacitance of an unknown Capacitor.

Theory

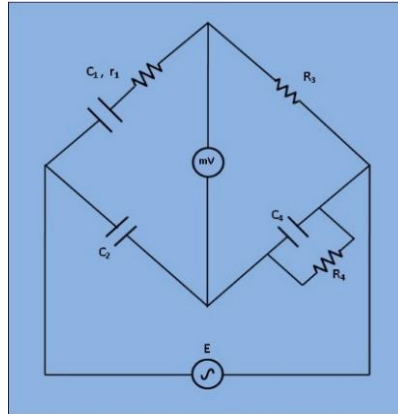


Fig 1: Circuit diagram for measurement of Capacitance by Schering Bridge

Let,

C_1 =capacitor whose capacitance is to be measured.

r_1 = a series resistance representing the loss in the capacitor C_1 .

C_2 = a standard capacitor.

R_3 = a non-inductive resistance.

C_4 = a variable capacitor.

R_4 = a variable non inductive resistance.

At balance,

$$\left(r_1 + \frac{1}{j\omega C_1}\right) * \left(\frac{R_4}{j\omega C_4 R_4 + 1}\right) = \frac{R_3}{j\omega C_2} \dots\dots (1)$$

$$r_1 R_4 - \frac{jR_4}{\omega C_1} = -\frac{jR_3}{\omega C_2} + \frac{R_3 R_4 C_4}{C_2} \dots\dots (2)$$

Or equating the real and imaginary terms in eq. (2), we obtain,

$$r_1 = R_3 * \frac{C_4}{C_2} \dots\dots (3)$$

$$C_1 = R_4 * \frac{C_2}{R_3} \dots\dots (4)$$

And, two independent balance equations (3) and (4) are obtained, if C_4 and R_4 are chosen as the variable elements.

Dissipation factor:

$$D_1 = \omega C_1 r_1 \dots (5)$$

Procedure:

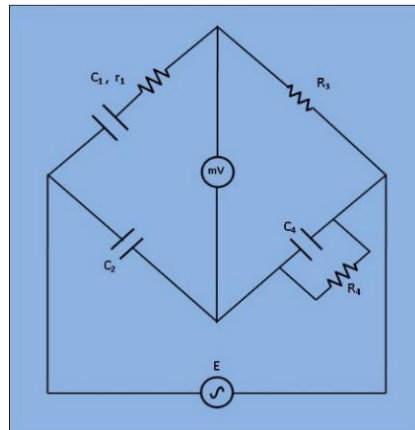
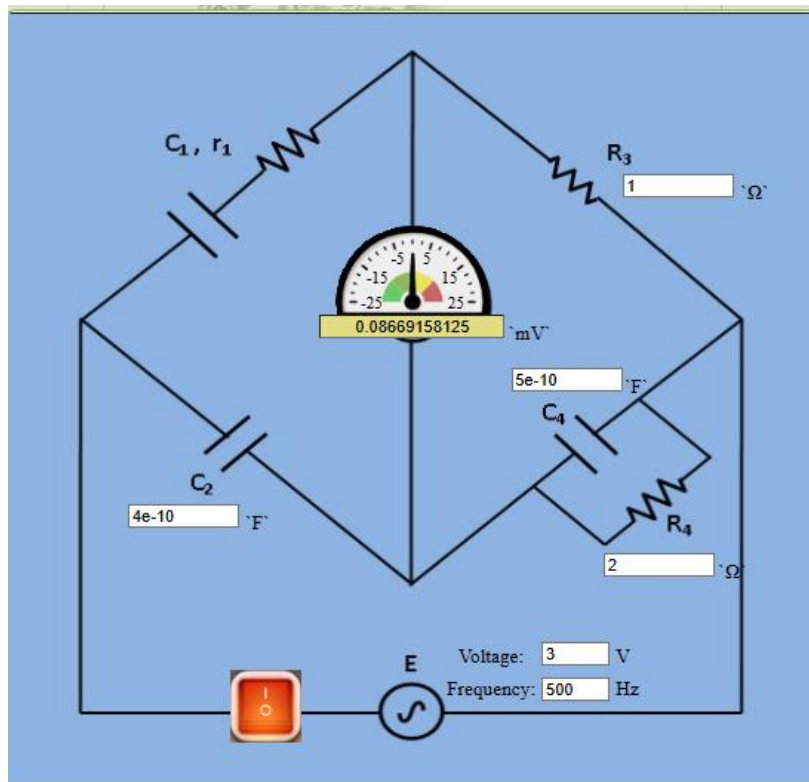


Fig. 1. Circuit diagram of experimental set-up for Capacitance measurement by Schering Bridge.

- 1) Apply Supply voltage from the signal generator with arbitrary frequency. ($V = 3V$). Also set the unknown Capacitance value from 'Set Capacitor Value' tab.
- 2) Then switch on the supply to get millivoltmeter deflection.
- 3) Choose the values of C_2 , C_4 , R_3 and R_4 from the capacitance and resistance box. vary the values to some particular values to achieve "NULL".
- 4) Observe the millivoltmeter pointer to achieve "NULL".
- 5) If "NULL" is achieved, switch to 'Measure Capacitor Value' tab and click on 'Simulate'. Observe the calculated values of unknown capacitance (C_1) and its internal resistance (r_1).
- 6) Also observe the Dissipation factor of the unknown capacitor which is defined as

$$\omega * C * r \text{ Where, } \omega = 2\pi f$$

Simulation:



The current voltmeter reading is: 0.0866915812 mv. Now simulate to get:

Capacitor value (in uF): 0.00080000

Resistance value (in Ω): 1.2500

Dissipation Factor: 0.0000031400

Simulate

RESULT:

Thus, the unknown capacitance is found using Schering Bridge.